

TO:

The Head of Department (HOD) & Infrastructure / Lab Administration

FROM:

Team OFFGRID (Finalist Team, Smart Horizon 48-Hour International Hackathon 2026)

WINDOW:

21st August 2026 → 1st September 2026 (11.5 Effective Working Days)

REQUISITION: DEDICATED SOFTWARE COLLABORATION WORKSPACE FOR MULTI-SUBSYSTEM SWARM ARCHITECTURE INTEGRATION

REMAINING PREP WINDOW

11.5 DAYS

Aug 21 PM → Sept 1 (Sept 2 Travel)

ARCHITECTURE SCALE

6 SUBSYSTEMS

Multi-UAV Autonomous Swarm

HACKATHON PRIZE POOL

₹23.75 LAKHS

Representing Institution Nationally

GRAND FINALE EVENT

SEPT 3-5

NHCE Campus, Bengaluru

01 PROJECT SUTRA: DEEP-TECH ARCHITECTURAL STRENGTHS & SCOPE

Developed by Team OFFGRID, Project SUTRA is an enterprise-grade autonomous multi-drone reconnaissance swarm engineered for disaster search and rescue in GPS-denied and communication-degraded environments. The system brings together 6 advanced, deep-tech research pillars:

AUTONOMOUS GNC & ORCA 3D

PX4 offboard 50Hz trajectory control, Visual-Inertial Odometry (VIO) for GPS-denied localization, and ORCA 3D reciprocal collision avoidance maintaining ≥3.5m clearance.

SWARMRAFT & DEEP JSCC

802.11s Wi-Fi mesh routing, SwarmRAFT distributed consensus (<100ms leader failover), and Deep JSCC neural image compression with PSNR ≥38.0 dB under severe RF jamming.

TRI-MODAL EDGE PERCEPTION

YOLOv8-Nano TensorRT inference (<5ms latency), Tri-Modal fusion (Visual, Thermal FLIR, mmWave Radar), and DEM raycast WGS84 target geolocator (<0.4m error).

02 TECHNICAL COMPLICATIONS & THE CRITICAL NEED FOR COLLOCATION

Integrating a 6-subsystem distributed robotics architecture across separate developer machines involves critical multi-threaded network bottlenecks that cannot be resolved asynchronously over chat:

SUBSYSTEM COUPLING	UNDERLYING TECHNICAL COMPLICATION	WHY IN-PERSON COLLOCATION IS MANDATORY
GNC ↔ Perception Loop (Subsystems A & C)	High-rate ROS 2 DDS camera feeds and 6-DoF pose transformation matrices must sync at 50Hz for real-time DEM target raycasting.	Simultaneous multi-monitor debugging of sensor pipelines and coordinate frame transforms side-by-side.
Swarm Mesh ↔ WebGPU GCS (Subsystems B & D)	WebSocket telemetry serialization under 5 live UAV streams requires tight synchronization to maintain 60.0 FPS HUD rendering.	Instant packet schema calibration and zero-latency WebSocket socket verification between frontend and backend leads.
11.5-Day Sprint Compression (All 6 Subsystems)	Asynchronous chat delays (2-3 hours per integration roadblock) create fatal schedule drift before the September 2nd travel deadline.	Instant 5-minute problem resolution, daily synchronized merge war-rooms, and unified end-to-end verification.

03

11.5-DAY SPRINT ROADMAP & DELIVERABLE COMMITMENTS

PHASE 1 • 4.0 DAYS

MODULE HARDENING

Aug 21 PM – Aug 24

- Resolve PyTorch / TensorRT runtime import conflicts.
- Sat, Aug 22:** Attend official NHCE Pre-Event Online Briefing.
- PX4 50Hz setpoints & SwarmRAFT consensus unit tests.
- WebGPU 3D GCS frontend WebSocket client initialization.

PHASE 2 • 4.0 DAYS

5-UAV SITL SWARM LOOP

Aug 25 – Aug 28

- GNC flight setpoints feed camera frames to Perception node.
- Perception publishes compressed target alerts over Mesh.
- GCS displays real-time 3D drone positions & 1-Click RTL.
- 5-Drone Gazebo Sim 8 digital twin run with ORCA 3D avoidance.

PHASE 3 • 3.5 DAYS

GATE AUDITS & FREEZE

Aug 29 – Sept 1 (18:00)

- Live Gate Audits G1–G6 verification against real pytest runs.
- Update D0CS .md strictly with measured statistics.
- 2x Timed Mock Presentations & Jury Defense Rehearsals.
- Sept 1, 18:00 IST:** Hard Code Freeze & 3x USB backup drives.

TEAM MEMBER	ASSIGNED SUBSYSTEM	SPECIFIC SOFTWARE MILESTONE (BY SEPT 1 FREEZE)	TEST VERIFICATION
Nikhil	SUBSYSTEMS A + B GNC, Comms & Sim	PX4 50Hz offboard setpoints, ORCA 3D collision avoidance, SwarmRAFT consensus, Gazebo Sim 8 digital twin.	pytest sutra_gnc/ pytest sutra_comms/
Vedanth	SUBSYSTEM C Edge Perception	YOLOv8-Nano TensorRT edge detector (<5ms), Tri-Modal fusion, DEM terrain raycast WGS84 target geolocator.	pytest sutra_perception/
Siva	SUBSYSTEM D 3D GIS GCS	WebGPU 60 FPS HUD, 3D Mapbox satellite drone tracking, WebSocket telemetry listener, Emergency 1-Click RTL.	npm run build (in sutra_gcs/)
Harika	SUBSYSTEM E Docs & Audits	Automated test suites, G1–G6 benchmark audit sync in D0CS .md, Master Pitch Deck visual formatting, PDF packages.	pytest full run & doc audit
Rohith	SUBSYSTEM F Tactical Ops	NDMA search patterns (Kedarnath flood / Wayanad landslide), Pre-flight SOP, operational rescue storytelling narrative.	CONOPS audit in docs/conops/

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WORKSPACE ALLOCATION REQUISITION

As registered finalists of **Team OFFGRID** representing the institution at the **Smart Horizon 48-Hour International Hackathon 2026 Grand Finale**, we formally requisition immediate administrative allocation of a dedicated software collaboration workspace for the duration of **21st August 2026 to 1st September 2026**. This space is required for intensive multi-subsystem swarm integration, real-time ROS 2 simulation verification, and end-to-end mission rehearsal prior to team departure.